

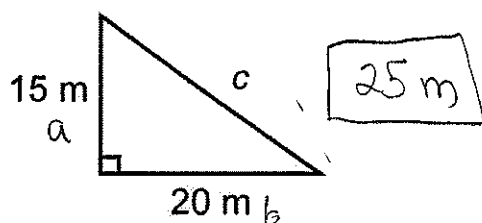
Chapter 8 Review

MPM1D

Part A: Pythagorean Theorem

1. Find the length of the unknown side of each triangle:

a)



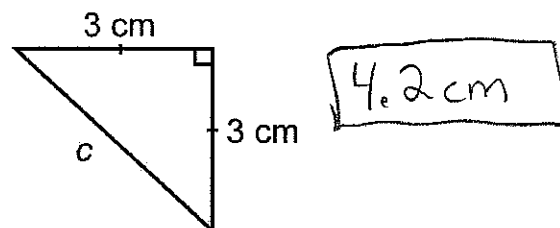
$$c^2 = a^2 + b^2$$

$$c^2 = 15^2 + 20^2$$

$$c^2 = 625$$

$$c = 25 \text{ m}$$

b)



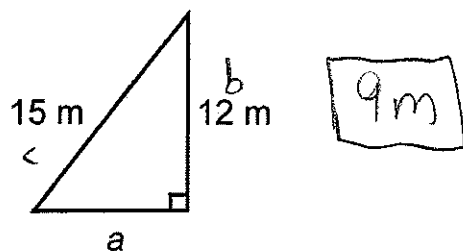
$$c^2 = a^2 + b^2$$

$$c^2 = 3^2 + 3^2$$

$$c^2 = 18$$

$$c = 4.2 \text{ cm}$$

c)



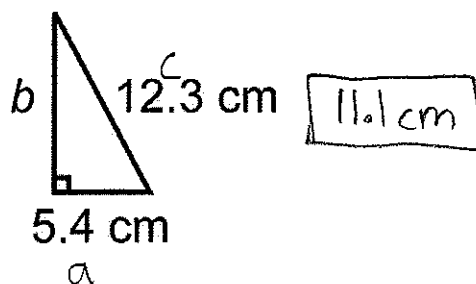
$$a^2 = c^2 - b^2$$

$$a^2 = 15^2 - 12^2$$

$$a^2 = 81$$

$$a = 9 \text{ m}$$

d)



$$b^2 = c^2 - a^2$$

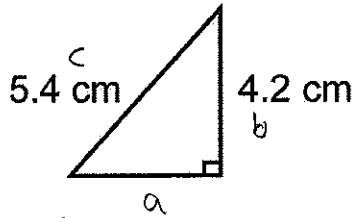
$$b^2 = 12.3^2 - 5.4^2$$

$$b^2 = 122.03$$

$$b = 11.1 \text{ cm}$$

2. Find the area and perimeter of each of the following triangles:

a)



$$a^2 = c^2 - b^2$$

$$a^2 = 5.4^2 - 4.2^2$$

$$a^2 = 11.62$$

$$a = 3.4 \text{ cm}$$

$$\text{Area} = \frac{1}{2}bh$$

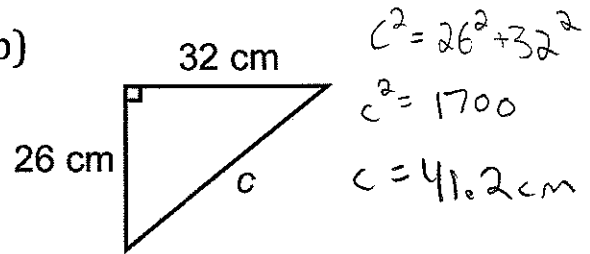
$$= \frac{1}{2}(3.4)(4.2)$$

$$= 7.14 \text{ cm}^2$$

$$\text{Perimeter} = 3.4 + 4.2 + 5.4$$

$$= 13 \text{ cm}$$

b)



$$c^2 = 26^2 + 32^2$$

$$c^2 = 1700$$

$$c = 41.2 \text{ cm}$$

$$A = \frac{1}{2}bh$$

$$= \frac{1}{2}(32)(26)$$

$$= 416 \text{ cm}^2$$

$$P = 26 + 32 + 41.2$$

$$= 99.2 \text{ cm}$$

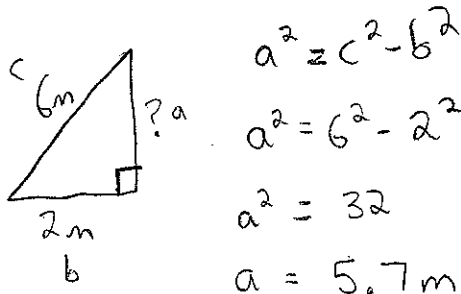
Area = 7.14 cm²

Perimeter = 13 cm

Area = 416 cm²

Perimeter = 99.2 cm

3. A 6-m extension ladder leans against a vertical wall with its base 2 m from the wall. How high up the wall does the top of the ladder reach? Round to the nearest tenth of a meter.



$$a^2 = c^2 - b^2$$

$$a^2 = 6^2 - 2^2$$

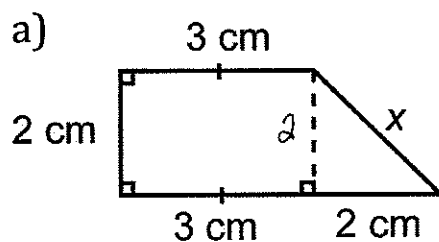
$$a^2 = 32$$

$$a = 5.7 \text{ m}$$

5.7m

Part B: Perimeter and Area of Composite Figures

4. Find the perimeter of each of the following figures:

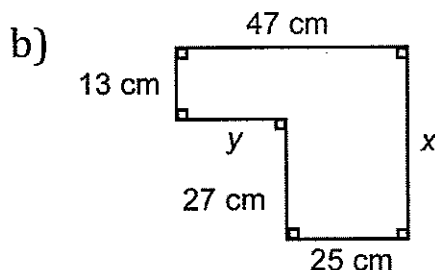


$$x^2 = 2^2 + 2^2$$

$$x^2 = 8$$

$$x = 2.8 \text{ cm}$$

$$\begin{aligned} \text{Perimeter} &= 2 + 3 + 2.8 + 2 + 3 \\ &= 12.8 \text{ cm} \end{aligned}$$

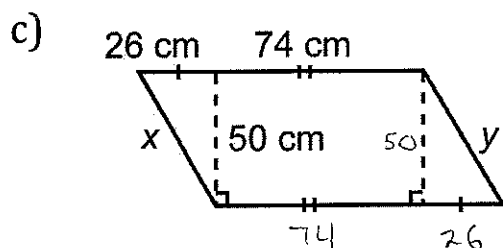


$$\begin{aligned} y &= 47 - 25 \\ &= 22 \text{ cm} \end{aligned}$$

$$x = 13 + 27$$

$$x = 40$$

$$\begin{aligned} \text{Perimeter} &= 47 + 40 + 25 + 27 + 22 + 13 \\ &= 174 \text{ cm} \end{aligned}$$

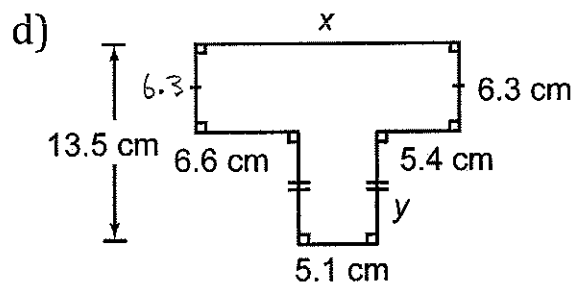


$$x^2 = 26^2 + 50^2$$

$$x^2 = 3176$$

$$x = 56.4 \text{ cm}$$

$$\begin{aligned} \text{Perimeter} &= 56.4 + 26 + 74 + 56.4 + 26 + 74 \\ &= 312.8 \text{ cm} \end{aligned}$$



$$x = 6.6 + 5.1 + 5.4$$

$$x = 17.1 \text{ cm}$$

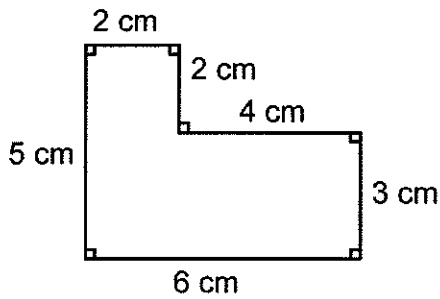
$$y = 13.5 - 6.3$$

$$y = 7.2 \text{ cm}$$

$$\begin{aligned} \text{Perimeter} &= 17.1 + 6.3 + 5.4 + 7.2 + 5.1 + 7.2 + 6.6 + 6.3 \\ &= 61.2 \text{ cm} \end{aligned}$$

5. Find the area of each of the following figures

a)

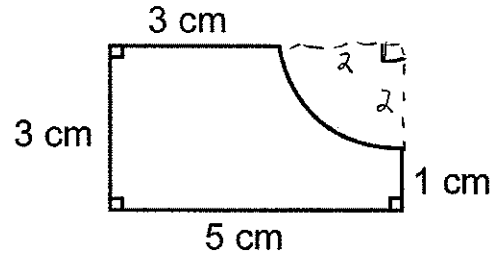


$$A_{\text{large}} = (5)(6) = 30 \text{ cm}^2$$

$$A_{\text{small}} = (2)(4) = 8 \text{ cm}^2$$

$$A_{\text{figure}} = 30 - 8 = 22 \text{ cm}^2$$

b) Hint: a quarter circle has been cut out

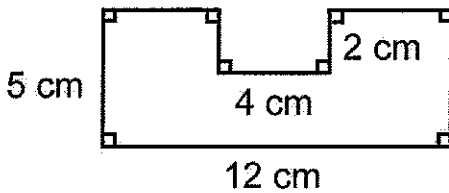


$$A_{\text{rectangle}} = (3)(5) = 15 \text{ cm}^2$$

$$A_{\frac{1}{4} \text{ circle}} = \frac{\pi(2)^2}{4} = 3.14 \text{ cm}^2$$

$$A_{\text{figure}} = 11.86 \text{ cm}^2$$

c)

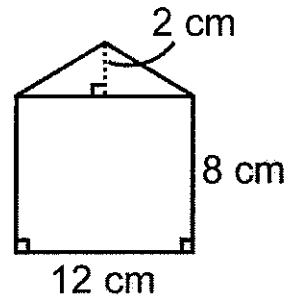


$$A_{\text{rectangle}} = (12)(5) = 60 \text{ cm}^2$$

$$A_{\text{cutout}} = (2)(4) = 8 \text{ cm}^2$$

$$A_{\text{figure}} = 60 - 8 = 52 \text{ cm}^2$$

d)

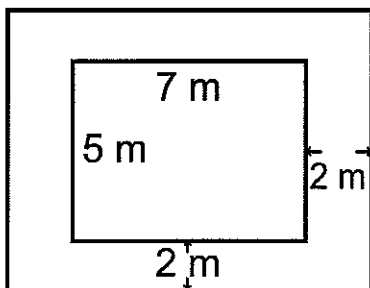


$$A_{\text{rectangle}} = (12)(8) = 96 \text{ cm}^2$$

$$A_{\text{triangle}} = \frac{1}{2}(12)(2) = 12 \text{ cm}^2$$

$$A_{\text{figure}} = 96 + 12 = 108 \text{ cm}^2$$

6. A pool is 5 m wide by 7 m long. It is surrounded by a deck 2 m wide. What is the area of the deck?



$$A_{\text{large rectangle}} = (9)(11) = 99 \text{ m}^2$$

$$A_{\text{small rectangle}} = (5)(7) = 35 \text{ m}^2$$

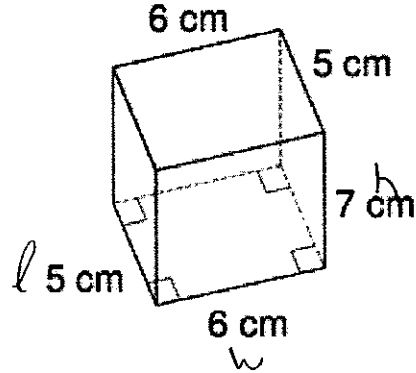
$$\text{Area of deck} = 99 - 35 = 64 \text{ m}^2$$

Part C: Prisms

7. For the following rectangular prism:

a) Find the volume:

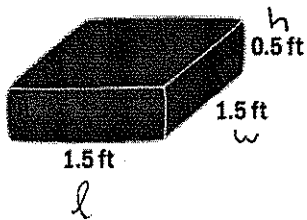
$$\begin{aligned} V &= lwh \\ &= (5)(6)(7) \\ &= 210 \text{ cm}^3 \end{aligned}$$



b) Find the surface area:

$$\begin{aligned} SA &= 2lw + 2wh + 2lh \\ &= 2(5)(6) + 2(6)(7) + 2(5)(7) \\ &= 60 + 84 + 70 \\ &= 214 \text{ cm}^2 \end{aligned}$$

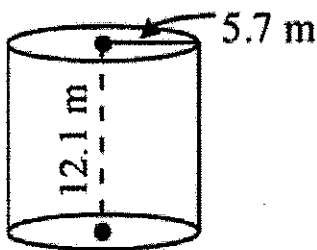
8. You are making the stadium cushion shown. The foam for the cushion costs \$1.50, and the fabric costs \$.50 per square foot. How much does it cost to make the cushion?



$$\begin{aligned} SA &= 2lw + 2wh + 2lh \\ &= 2(1.5)(1.5) + 2(1.5)(0.5) + 2(1.5)(0.5) \\ &= 4.5 + 1.5 + 1.5 \\ &= 7.5 \end{aligned}$$

$$\begin{aligned} \text{Cost} &= 1.50 + 7.5(0.5) \\ &= \$5.25 \end{aligned}$$

9. Find the volume and surface area of the following cylinder:



$$\begin{aligned} V &= \pi r^2 h \\ &= \pi (5.7)^2 (12.1) \\ &= 1235.1 \text{ m}^3 \end{aligned}$$

$$\begin{aligned} SA &= 2\pi r^2 + 2\pi rh \\ &= 2\pi (5.7)^2 + 2\pi (5.7)(12.1) \\ &= 637.5 \text{ m}^2 \end{aligned}$$

10. A cylindrical tank has a diameter of 2.60 m and a height of 1.65 m.

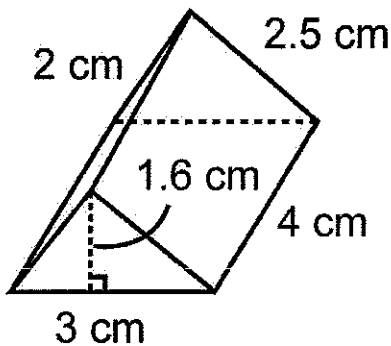
a) Determine the volume of the tank

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi (1.3)^2 (1.65) \\ &= 8.8 \text{ m}^3 \end{aligned}$$

b) Express the storage capacity in Litres. (Remember $1 \text{ m}^3 = 1000 \text{ L}$)

$$8.8 \text{ m}^3 \times \frac{1000 \text{ L}}{1 \text{ m}^3} = 8800 \text{ L}$$

11. Find the volume and Surface area of the following triangular prism:



$$\begin{aligned} V &= \frac{1}{2} b l h \\ &= \frac{1}{2} (3) (1.6) (4) \\ &= 9.6 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Area of triangle} &= \frac{1}{2} b h \\ &= \frac{1}{2} (3) (1.6) \\ &= 2.4 \text{ cm}^2 \end{aligned}$$

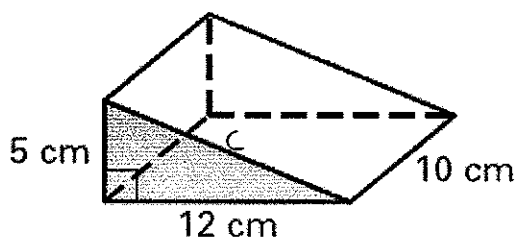
$$\begin{aligned} \text{Area rectangle ①} &= (2.5) (4) \\ &= 10 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area rectangle ②} &= (2) (4) \\ &= 8 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area rectangle ③} &= (3) (4) \\ &= 12 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} SA &= 2(2.4) + 10 + 8 + 12 \\ &= 34.8 \text{ cm}^2 \end{aligned}$$

12. Here is a picture of a block of cheese Mr. Jensen had for breakfast.



$$c^2 = 5^2 + 12^2$$

$$c^2 = 169$$

$$c = 13$$

a) Mr. Jensen had to remove the package before eating it. How much packaging did he remove? (Find the surface area)

$$A_{\text{triangle}} = \frac{1}{2}(5)(12) \\ = 30 \text{ cm}^2$$

$$A_{\text{rectangle ①}} = (13)(10) \\ = 130 \text{ cm}^2$$

$$A_{\text{rectangle ②}} = (12)(10) \\ = 120 \text{ cm}^2$$

$$A_{\text{rectangle ③}} = (5)(10) \\ = 50 \text{ cm}^2$$

$$SA = 2(30) + 130 + 120 + 50 \\ = 360 \text{ cm}^2$$

b) How many cubic centimeters of cheese is in Mr. Jensen's stomach right now? (what is the volume of the triangular prism?)

$$V = \frac{1}{2}bhl$$

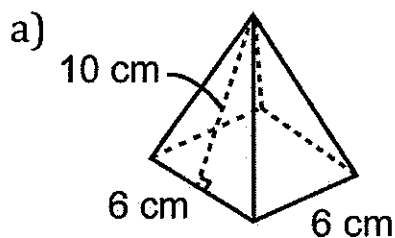
$$= \frac{1}{2}(12)(5)(10)$$

$$= 300 \text{ cm}^3$$

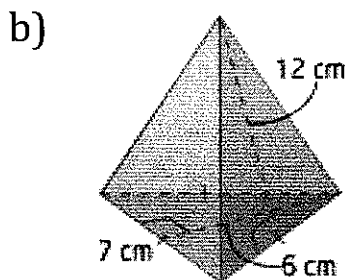
Part D: Pyramids

13.

13. Find the surface area of the following pyramids:



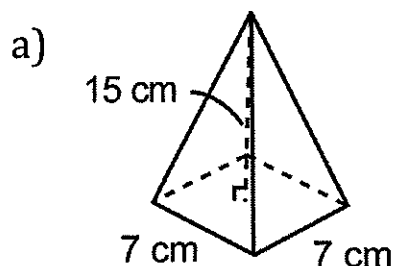
$$\begin{aligned} SA &= (6)(6) + 4\left(\frac{1}{2}\right)(6)(10) \\ &= 36 + 120 \\ &= 156 \text{ cm}^2 \end{aligned}$$



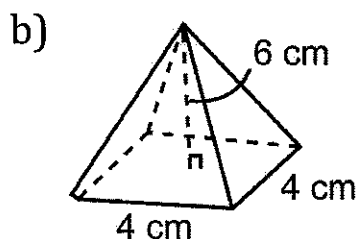
$$\begin{aligned} SA &= \frac{1}{2}(7)(6) + 3\left(\frac{1}{2}\right)(7)(12) \\ &= 21 + 126 \\ &= 147 \text{ cm}^2 \end{aligned}$$

14.

14. Find the volume of the following pyramids



$$\begin{aligned} V &= \frac{1}{3}(7)(7)(15) \\ &= 245 \text{ cm}^3 \end{aligned}$$



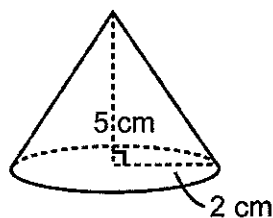
$$\begin{aligned} V &= \frac{1}{3}(4)(4)(6) \\ &= 32 \text{ cm}^3 \end{aligned}$$

Part E: Cones

15.

15. Find the volume of the following cones:

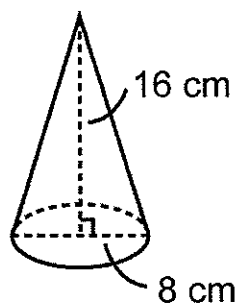
a)



$$V = \frac{1}{3} \pi (2)^2 (5)$$

$$= 20.9 \text{ cm}^3$$

b)



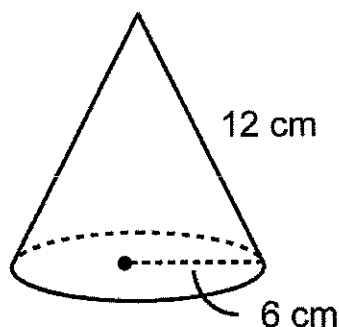
$$V = \frac{1}{3} \pi (8)^2 (16)$$

$$= 268.1 \text{ cm}^3$$

16.

16. Find the surface area of the following cones:

a)

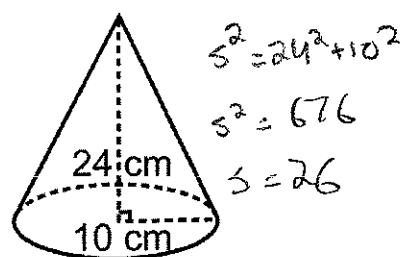


$$SA = \pi r s + \pi r^2$$

$$= \pi (6)(12) + \pi (6)^2$$

$$= 339.3 \text{ cm}^2$$

b)



$$s^2 = 24^2 + 10^2$$

$$s^2 = 676$$

$$s = 26$$

$$SA = \pi (10)(26) + \pi (10)^2$$

$$= 1131.0 \text{ cm}^2$$

17

17. A cone has volume 424.1 cm^3 and height 5 cm. What is the base radius of the cone?

$$V = \frac{1}{3} \pi r^2 h$$

$$424.1 = \frac{1}{3} \pi r^2 (5)$$

$$1272.3 = \pi r^2 (5)$$

$$r^2 = 81$$

$$r = 9$$

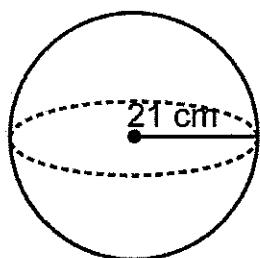
9 cm

Part F: Spheres

18.

Find the volume and surface area of the following spheres

a)



$$V = \frac{4\pi r^3}{3}$$

$$= \frac{4\pi (21)^3}{3}$$

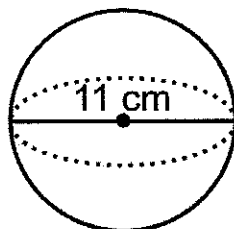
$$= 38792.4 \text{ cm}^3$$

$$SA = 4\pi r^2$$

$$= 4\pi (21)^2$$

$$= 5541.8 \text{ cm}^2$$

b)



$$V = \frac{4\pi (5.5)^3}{3}$$

$$= 696.9 \text{ cm}^3$$

$$SA = 4\pi (5.5)^2$$

$$= 380.1 \text{ cm}^2$$

19. Find the radius of a sphere that has Surface Area = 452.4 cm^2

$$SA = 4\pi r^2$$

$$452.4 = 4\pi r^2$$

$$r^2 = 36$$

$$r = 6$$

$$16 \text{ cm}$$

20.

A sphere just fits inside a cube with edges 18 cm long.

a) Find the volume of the sphere. $V = \frac{4\pi r^3}{3}$

$$= \frac{4\pi (9)^3}{3}$$

$$= 3053.6 \text{ cm}^3$$

b) What is the volume of the cube?

$$V = (18)^3$$

$$= 5832 \text{ cm}^3$$

c) How much empty space is there in the cube?

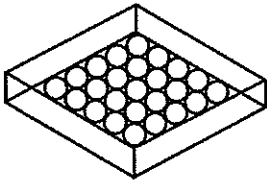
$$= 5832 - 3053.6$$

$$= 2778.4 \text{ cm}^3$$

Part G: Application

21)

Twenty-five balls, each with diameter 9 cm, are packed in a single layer in a square box.



a) What is the minimum volume of the box?

$$\begin{aligned}\text{Length} &= (5)(9) \\ &= 45 \text{ cm}\end{aligned}$$

$$\text{height} = 9 \text{ cm}$$

$$\begin{aligned}V &= (45)(45)(9) \\ &= 18225 \text{ cm}^3\end{aligned}$$

b) What is the surface area of the box?

$$\begin{aligned}SA &= 2(45)(45) + 2(45)(9) + 2(45)(9) \\ &= 5670 \text{ cm}^2\end{aligned}$$

c) How much empty space is in the box?

$$\begin{aligned}\text{Volume of a ball} &= \frac{4\pi r^3}{3} \\ &= \frac{4\pi (4.5)^3}{3} \\ &= 381.7 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Empty space} &= 18225 - 25(381.7) \\ &= 8682.5 \text{ cm}^3\end{aligned}$$